

Artificial Intelligence

Concise Study Notes: Chapters 1, 6 and 7

ENCT 351 Year III, Part II

Scope. These notes follow the syllabus and micro syllabus for Chapter 1 (Introduction, 4 hours), Chapter 6 (AI Applications, 5 hours) and Chapter 7 (Emerging Trends, 2 hours), which together carry about 14 of the 60 theory marks. Topics marked with an asterisk in the micro syllabus as descriptive only are left out, except where the model question paper asks about them. Forty MCQs with an answer key are at the end.

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Chapter 1 Introduction

1.1 Definition, Foundation and History of AI

Definitions

Intelligence is the ability to process information to solve problems and adapt to new situations.

Artificial Intelligence (AI) is “the science and engineering of making intelligent machines, especially intelligent computer programs” (John McCarthy). Equivalently, it is the capability of computational systems to perform tasks normally associated with human intelligence: learning, reasoning, problem solving, perception and decision making. The term was coined for the Dartmouth conference of summer 1956 (McCarthy, Minsky, Rochester, Shannon).

Four approaches to AI

Researchers have pursued AI along two axes: whether the goal is to match *human* performance or an ideal *rational* standard, and whether the focus is on *thought* or on *behaviour*.

	Human like	Rational
Thought	Thinking humanly: cognitive modelling. Build programs that think as the brain does, using introspection, psychological experiments and brain imaging.	Thinking rationally: “laws of thought” (Aristotle’s syllogisms). Correct inference from premises using logic.
Behaviour	Acting humanly: the Turing test approach.	Acting rationally: the rational agent approach. Act to achieve the best (expected) outcome.

Problems. Cognitive modelling is hard because we do not fully understand how humans think, and machines do not have brain hardware. The laws of thought approach faces the *translation problem* (informal, uncertain knowledge is hard to state in strict logic) and the *computational problem* (even a few dozen facts can give trillions of possibilities). The **rational agent approach is the most appropriate**: it is scientifically measurable through a performance measure and it handles uncertainty. Its limit is that perfect rationality is computationally unattainable.

Turing test, Total Turing test and Reverse Turing test

Alan Turing (1950) asked “Can machines think?” and proposed an imitation game. A human interrogator exchanges written questions with two hidden parties, one human and one machine. If the interrogator cannot tell which is which, the machine is said to be intelligent.

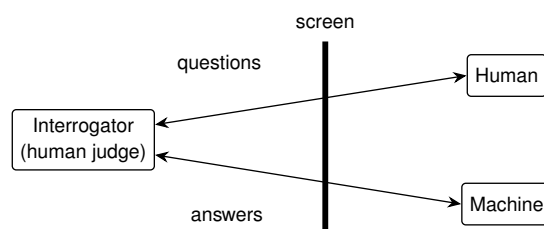


Figure 1: The Turing test set up. The judge sees only text.

- A machine passing the **Turing test** needs four capabilities: *natural language processing* (to communicate), *knowledge representation* (to store what it knows and hears), *automated reasoning* (to answer using stored information) and *machine learning* (to adapt to new circumstances).
- The **Total Turing test** adds a physical, perceptual channel. The machine also needs *computer vision and speech recognition* to perceive, and *robotics* to manipulate objects and move.
- The **Reverse Turing test** flips the roles: a machine judges whether the other party is a human. The standard example is a CAPTCHA, used for security as “proof of personhood”.
- *Criticism:* the test rewards imitation rather than understanding. In Loebner Prize contests, judges sometimes labelled real humans as machines, showing the weakness of the method.

Foundations of AI

Discipline	Contribution to AI
Philosophy	Logic, reasoning, mind as a physical system, foundations of learning, language and rationality.
Mathematics	Formal representation and proof, computation, undecidability, intractability, probability.
Economics	Formal theory of rational decisions, game theory, operations research (maximising payoff).
Neuroscience	The physical substrate of mental activity: how brains process information.
Psychology	Adaptation, perception and motor control: how humans and animals think and act.
Linguistics	Knowledge representation and grammar: how language relates to thought.
Computer engineering	Building efficient computers to run intelligent programs.

History of AI

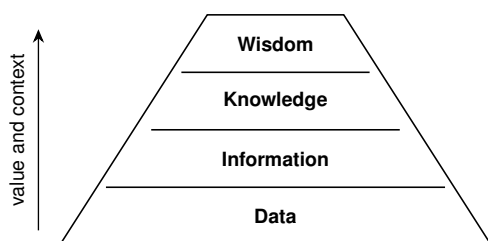
Year	Milestone
1946	ENIAC, the first programmable, general purpose electronic digital computer (not intelligent itself, but the hardware base).
1950	Turing publishes “Computing Machinery and Intelligence” and proposes the Turing test.
1956	Dartmouth conference: AI is born as a field.
1964, 1966	ELIZA, the first chatbot psychotherapist. Shakey, the first general purpose mobile robot (Stanford), which led to the A* search algorithm.
1974 to 1980	First AI winter: machine translation criticised, funding cuts, negative results on neural networks (XOR problem).
1986	Backpropagation popularised (Rumelhart, Hinton, Williams), enabling multilayer networks, but data and computers were still inadequate.
1987 to 1993	Second AI winter: expert systems could not learn and needed costly human experts.
1997	IBM Deep Blue defeats world chess champion Garry Kasparov.
1998	LeNet, a convolutional network trained with backpropagation, reads handwritten digits.
2005, 2007	DARPA Grand Challenge won by Stanford’s Stanley (probabilistic robotics); Urban Challenge won by CMU’s Tartan Racing.
2012	AlexNet wins ImageNet using GPUs: big data + fast GPUs + neural networks gives superhuman perception.
2016	DeepMind’s AlphaGo defeats Go champion Lee Sedol.
2017	“Attention Is All You Need” introduces the Transformer.
2022 to 2023	Generative AI: ChatGPT, DALL E 2 and large language models.
Present	Multimodal models, AI agents, and “world models” for spatial intelligence.

1.2 Definition and Importance of Knowledge

Knowledge

Knowledge is a theoretical or practical understanding of a subject. Formally, it is “information combined with experience, context, interpretation and reflection, a high value form of information that is ready to apply to decisions and actions” (Davenport et al., 1998). It is also described as human expertise stored in a person’s mind, gained through experience and interaction with the environment.

Importance. (i) *Speed and efficiency*: knowledge lets us skip trial and error, for example knowing which berries are poisonous. (ii) *Problem solving*: one cannot solve a problem without understanding its variables. (iii) *Innovation*: new ideas are mostly new combinations of existing knowledge. In AI, every reasoning system (an expert system, a knowledge based agent) works only as well as the knowledge it holds. Learning is the process that acquires knowledge: knowledge is the destination, learning is the journey.



Level	Meaning and example
Data	Raw facts: 34, 36, 38 (daily temperatures).
Information	Data organised to be meaningful: “average temperature this week is 36 degrees”.
Knowledge	Understanding from analysis: “above 35 degrees, heat stress is likely for crops”.
Wisdom (intelligence)	Sound judgment: “irrigate at night and shift planting dates”.

Figure 2: The DIKW pyramid.

1.3 Cognition and Learning (Neuroscience)

Neuroscience view. The brain is a network, not a single muscle. It has about 86 billion neurons, each linked to thousands of others through **synapses**. Intelligence arises from the connections and the speed and efficiency of electrical signals along them. When we learn, the brain physically rewires itself by strengthening connections between particular neurons. This idea is the inspiration for artificial neural networks.

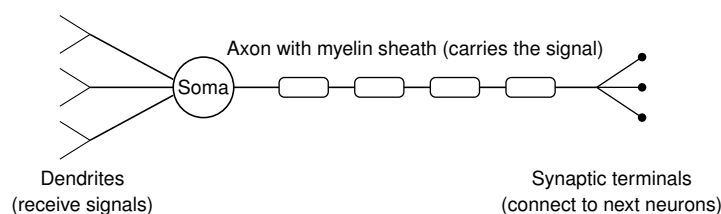
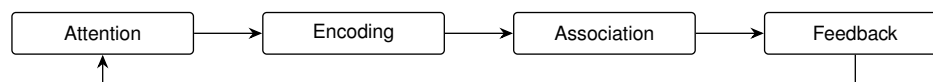


Figure 3: A biological neuron.

Brain part	Role
Neocortex	Logic, language, planning, mathematics (what IQ tests measure). The layered structure inspired deep learning.
Cerebellum	Movement, balance, muscle memory.
Limbic system	Emotion, fear, motivation.
Conscious mind	Focused thinking, willpower, logic. Subconscious mind: autopilot functions (breathing, habits, emotional reactions), stores all memories, source of intuition.

Human cognition. Cognition is the mental process of acquiring knowledge and understanding through thought, experience and the senses: thinking, knowing, remembering, judging and problem solving. To model it, we use introspection, psychological experiments and brain imaging.

Learning is the active process of acquiring new understanding, knowledge, behaviours, skills, values and preferences. It involves four elements:



Attention (you cannot learn what you ignore), encoding (translating information into a usable form), association (linking new data to existing mental frameworks) and feedback (observing success or failure and adjusting).

Cognitive computing refers to systems that mimic human thought processes in a computerised model, by learning from data, reasoning about it and interacting naturally with people. Typical uses: healthcare diagnostics, financial analysis and customer service.

State of the art. “State of the art” means the highest level of development reached at a given time. Representative areas:

Field	Current level
Robotics and vehicles	Supervised self driving on public roads and driverless robotaxis (for example Waymo); humanoid and legged robots moving from research to pilot use.
Medicine and science	AlphaFold predicts protein structures (Nobel Prize in Chemistry, 2024); AI matches or exceeds specialists on many diagnostic imaging tasks.
Language and creativity	Reasoning capable large language models; text to image and text to video generation (for example Sora, Veo).
Environment	AI weather models such as GraphCast forecast extreme weather faster than physics based supercomputer models.
Software engineering	From code autocomplete to autonomous coding agents that read an issue, edit a codebase and run tests.

1.4 Human Intelligence and Machine Intelligence

AI is a *specialist engine*: it excels at processing massive data, executing rules and scaling output. Humans are *generalist architects*: they handle ambiguity, ethical judgment and context, and they manage the AI itself.

Parameter	Human	Machine (AI)	Better
Speed and scale	Slow and linear; limited reading and writing rate.	Fast and massive; processes millions of words quickly.	AI
Reasoning	Deep and novel; invents new theories, but slow at calculation.	Strong on structured maths and science problems.	Both
Creativity	Intentional and emotional, from lived experience.	Generative, but largely remixes learned patterns.	Human
Physical world	Highly adaptable in messy environments.	Improving but rigid; struggles when surroundings change.	Human
Reliability	Varies with fatigue and attention.	Consistent once trained; does not tire.	AI
Adaptability	Universal: learns chess, cooking and driving in one lifetime.	Brittle: a chess AI cannot cook; general models struggle when rules change.	Human

1.5 AI Tree: Branches and Interdisciplinary Nature

AI is a broad field. Its main branches, each with typical applications, are shown below.

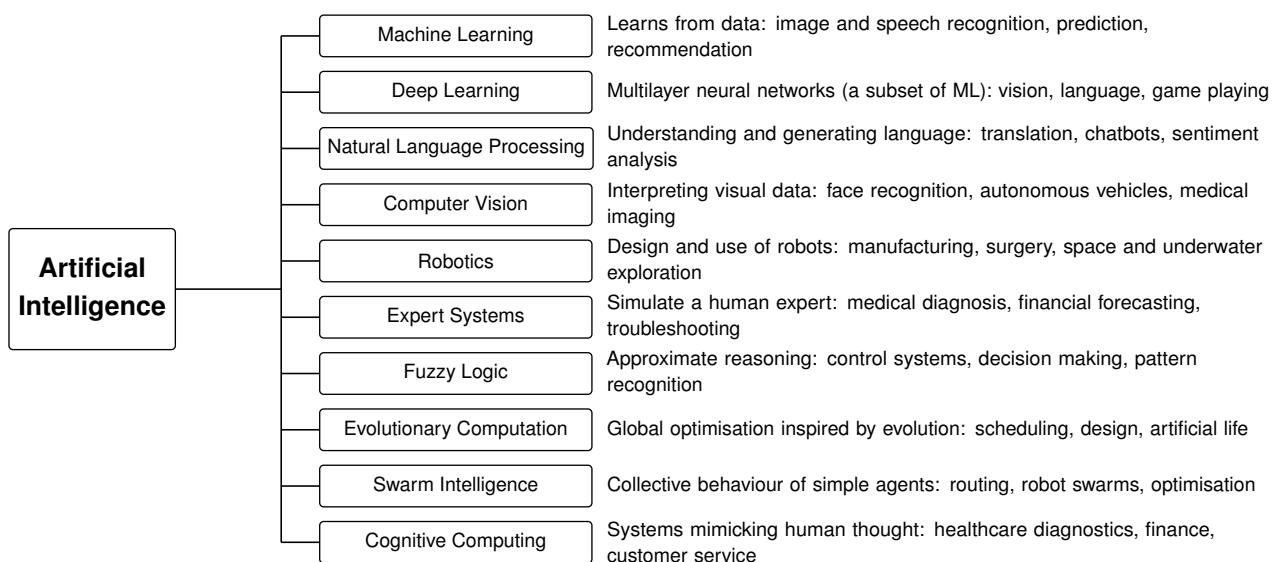


Figure 4: Branches of the AI tree with typical applications.

Interdisciplinary applications.

Domain	Uses of AI
Healthcare	Medical image analysis (X rays, MRI, CT), predicting patient outcomes, drug discovery, personalised medicine.
Finance	Real time fraud detection, algorithmic trading, chatbots for customer service.
Autonomous vehicles	Perception, decision making and control for self driving cars; adaptive traffic signal management.
Retail and online shopping	Recommendation (collaborative filtering uses user similarity; content based filtering uses item features), demand forecasting, support chatbots.
Manufacturing	Predictive maintenance, visual quality inspection, supply chain optimisation.
Language (NLP)	Translation, sentiment analysis, voice assistants.
Education	Personalised learning, automated grading, virtual tutors.
Entertainment, law, others	Content generation and editing; contract review and case outcome prediction; agriculture, marketing, governance, energy, human resources.

1.6 Intelligent Agents and Types

Agent and rational agent

An **agent** is anything that perceives its environment through *sensors* and acts upon it through *actuators*. A **percept** is the agent's input at one instant; the **percept sequence** is the complete history. The **agent function** maps percept sequences to actions, $f : P^* \rightarrow A$, and the agent program implements it. An **intelligent agent** is an agent that perceives, decides and acts to achieve goals, adapting to its environment.

A **rational agent** selects the action expected to maximise its performance measure, given the percept sequence and its built in knowledge:

$$a^* = \arg \max_a \mathbb{E}[\text{performance measure} \mid \text{percept sequence}, a].$$

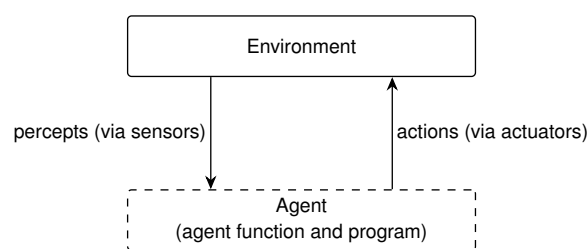


Figure 5: An agent interacts with its environment through sensors and actuators.

Applications: autonomous systems, personal assistants, recommendation systems, financial trading. **Challenges:** ethical and legal implications, robustness and reliability, interpretability, scalability and efficiency.

PEAS framework (task environment specification)

To design an agent, specify its **P**erformance measure, **E**nvironment, **A**ctuators and **S**ensors.

Agent	Performance	Environment	Actuators	Sensors
Vacuum cleaner	Cleanliness, battery use, time	Rooms, floor, dirt, furniture	Wheels, brushes, suction	Dirt sensor, bump sensor, camera
Self driving taxi	Safety, speed, comfort, legality	Roads, traffic, pedestrians	Steering, brake, accelerator, horn	Cameras, lidar, GPS, speedometer
Chess player	Win, time control	Board, opponent	Move output	Board state input

Factors to consider when selecting an agent

- **Environment:** fully or partially observable; deterministic or stochastic; static or dynamic (does the world change while the agent thinks?); episodic or sequential; discrete or continuous; single or multi agent.
- **Task complexity:** need for long term planning (chess) versus isolated decisions (labelling a photo); a simple pass or fail goal versus trade offs (speed versus safety); working alone versus competing or cooperating.
- **Resource constraints:** reaction time (reflex in milliseconds versus deliberate planning), memory available for history, battery or power limits.
- **Adaptability:** must it improve itself over time, tolerate a failing sensor, handle brand new situations?

Types of agents

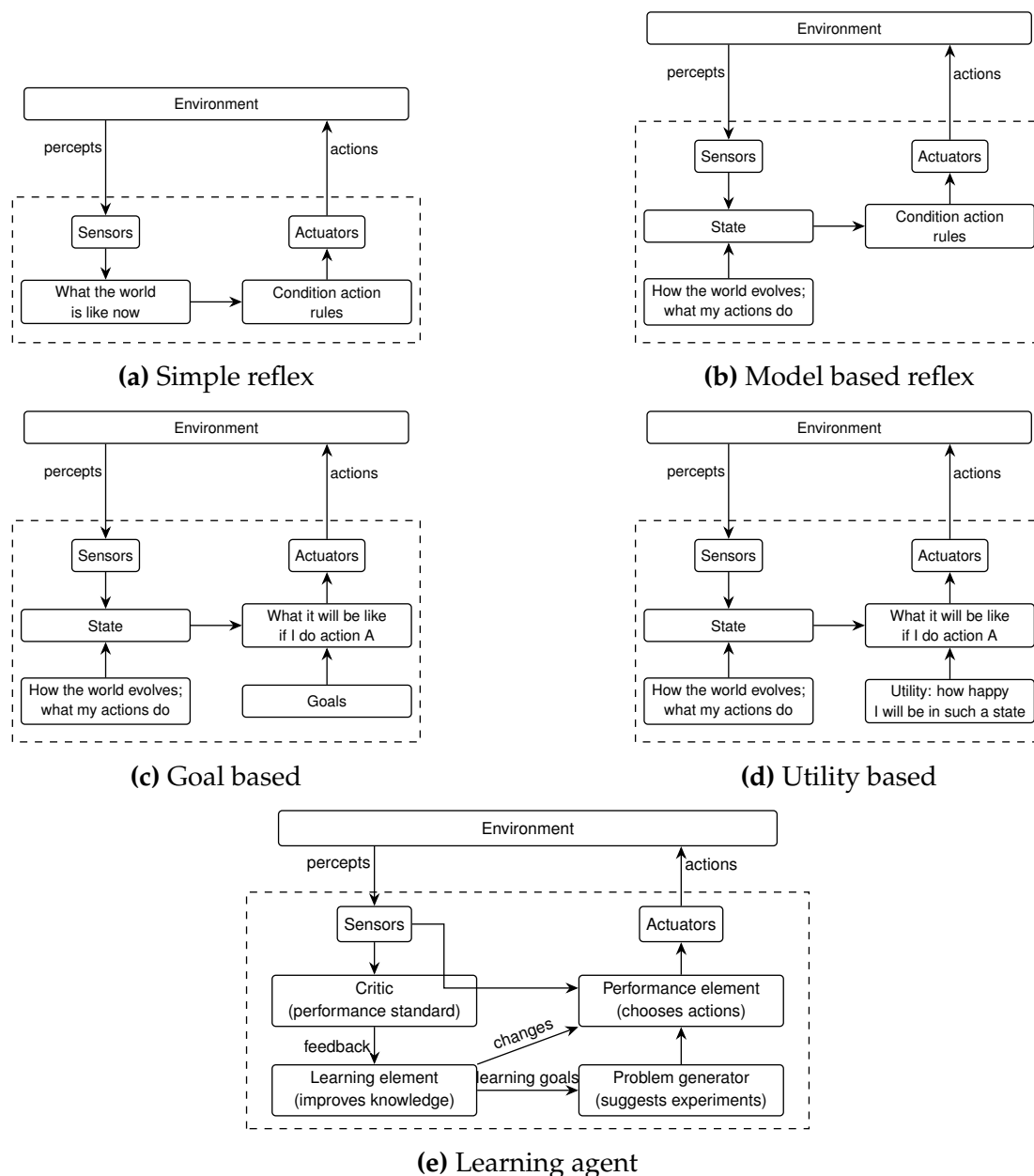


Figure 6: Structures of the main agent types. The dashed box is the agent.

Type	Idea	Strengths and weaknesses	Example
Simple reflex	Acts on the current percept only, using condition action rules.	Fast, simple; fails if history matters or the world is partially observable.	Timer based traffic light.
Model based reflex	Keeps an internal state (a model of how the world evolves).	Handles partial observability; needs more memory and computation.	Robot vacuum that maps rooms and remembers cleaned areas.
Goal based	Chooses actions by considering future states against explicit goals, using search and planning.	Flexible, can re plan; slower.	Delivery route planner.

Type	Idea	Strengths and weaknesses	Example
Utility based	Maximises a utility function, so it reaches goals in the <i>best</i> way and trades off conflicting objectives.	Handles uncertainty and trade offs; utility functions are hard to define.	Investment portfolio agent.
Learning	Improves through a critic, learning element, performance element and problem generator.	Adapts to new environments; needs data and time.	Chatbot that improves with feedback.
Multi agent	Several agents share an environment, cooperating or competing; needs communication and coordination.	Suits decentralised problems; coordination is complex.	Traffic control, robot teams.
Hierarchical	Layers of strategic, tactical and operational control; high levels set subgoals for lower ones.	Breaks complex tasks into simpler ones.	Drone delivery: fleet level plus individual navigation.

1.7 Agentic AI

Agentic AI

Agentic AI refers to AI systems that show *agency*: they pursue complex goals autonomously by breaking them into subtasks, choosing and using external tools, observing the results and correcting themselves, with limited human supervision.

Feature	Earlier (generative) AI	Agentic AI
Primary function	Generates text or images.	Executes tasks and workflows.
Interaction	Conversation (prompt and response).	Given a goal, then works towards it.
Environment access	Mostly read only.	Read and write: can change the state of the world (files, apps, orders).
On failure	Hallucinates or apologises.	Self corrects and tries another approach.

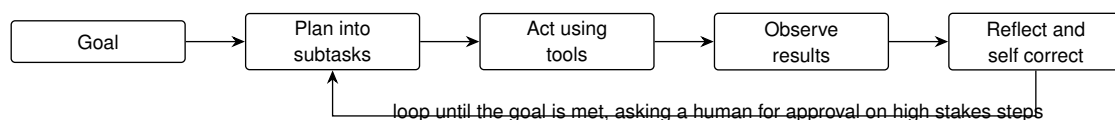


Figure 7: The agentic loop.

Characteristics.

- **Proactive:** does not wait to be told; anticipates problems from historical data and trends and acts before they escalate.
- **Adaptable:** understands the rules and tone of its domain, changes strategy in real time, and handles edge cases that rigid scripted software would reject.
- **Collaborative:** works as a copilot, asks for human approval on high stakes decisions, and shares status with other agents.

- **Specialised:** focuses on being excellent at one task, and splits a large goal into subtasks routed to the agents best suited to them.

Autonomous intelligence is the highest level of AI maturity: systems that operate independently in open ended environments to achieve high level objectives without constant human guidance. Examples: NASA's Perseverance rover, Amazon Proteus warehouse robots, Waymo robotaxis.

How intelligent agents and agentic AI lead to autonomous intelligence. An intelligent agent supplies the basic perceive, decide and act cycle. Agentic AI extends it with goal decomposition, tool use, memory and self correction, so that the agent can pursue an objective over many steps. Together they let a system sense a changing environment, plan, act and recover from errors on its own, which is what autonomous intelligence requires.

Chapter 6 AI Applications

6.1 Expert Systems: Characteristics, Architecture, Development and Applications

Expert system

An **expert system** is an AI program that simulates the decision making ability of a human expert in a narrow domain. It stores domain knowledge as facts and rules in a *knowledge base* and applies an *inference engine* to solve problems and give advice. The key idea is to preserve and replicate human expertise.

Characteristics.

- *Expert level performance* in a specific, narrow domain (medicine, chemistry, fault diagnosis).
- *Knowledge is separate from control*: rules live in the knowledge base; reasoning lives in the inference engine, so knowledge can be changed without rewriting the program.
- *Symbolic reasoning* with rules of the form IF condition THEN conclusion.
- *Explanation ability*: it can tell the user how and why it reached a conclusion.
- *Consistent and available*: gives the same answer for the same facts and works at any time. It can also handle uncertain or incomplete information (for example with certainty factors).

Importance: preserves expertise when experts retire, gives consistent and unbiased advice, saves time and cost by automating expert work, and makes expert knowledge accessible to non experts.

Architecture and components

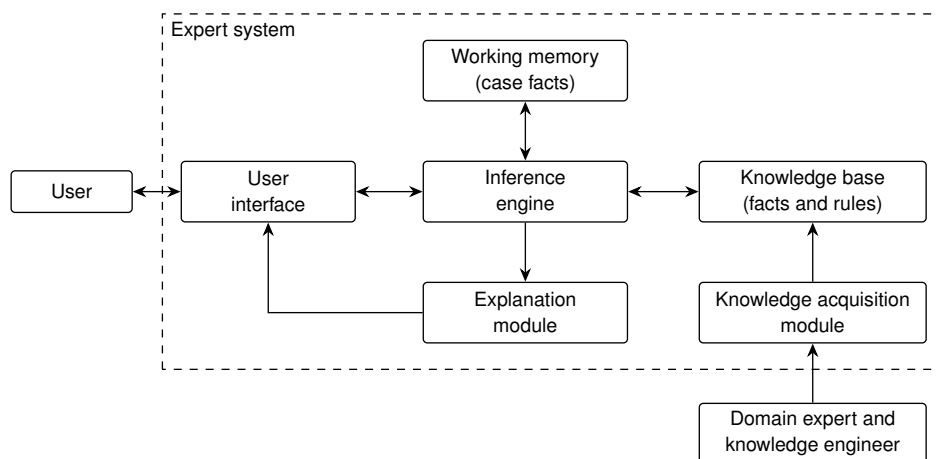


Figure 8: Architecture of an expert system.

- **Knowledge base:** the heart of the system; domain facts and rules, for example IF fever AND cough THEN suspect flu.
- **Inference engine:** the brain; applies rules to facts using *forward chaining* (data driven: from known facts to conclusions) or *backward chaining* (goal driven: from a hypothesis back to the facts that support it). The facts of the current case are held in *working memory*.
- **User interface:** lets a non expert enter a query and read the answer.

- **Explanation module:** shows the step by step reasoning behind a conclusion (“why do you ask?”, “how did you conclude that?”), which builds trust.
- **Knowledge acquisition module:** updates and extends the knowledge base so the system does not become outdated.

Pros and cons.

Benefits	Limitations
<i>Consistency:</i> same facts give same advice.	<i>Knowledge limitation:</i> only knows its narrow domain, no common sense.
<i>Availability:</i> works at any time without fatigue.	<i>Lack of flexibility:</i> cannot learn by itself or handle cases outside its rules.
<i>Cost effectiveness:</i> reduces cost of hiring experts.	<i>Maintenance:</i> keeping rules current is slow and expensive.
<i>Knowledge preservation:</i> keeps expertise within the organisation.	Acquiring knowledge from experts is difficult (the knowledge acquisition bottleneck).

Development phases

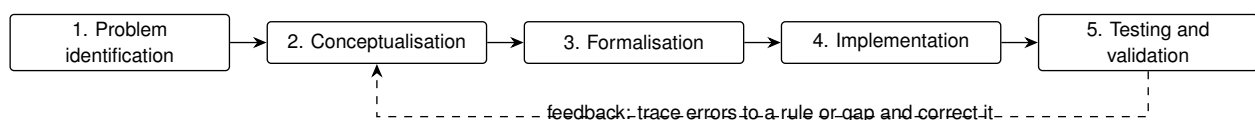


Figure 9: The five development phases of an expert system.

1. **Problem identification:** decide whether an expert system is the right tool; the problem must be narrow enough to manage but complex enough to need real expertise.
2. **Conceptualisation:** the knowledge engineer works closely with the domain expert to uncover strategies, rules of thumb and constraints, and models the expert’s thinking without coding.
3. **Formalisation:** translate the concepts into a representation such as IF THEN rules, frames or semantic networks.
4. **Implementation:** build the knowledge base, configure the inference engine and create the user interface to get a working prototype.
5. **Testing and validation:** run real cases and compare the system’s answers with the expert’s. Errors are traced to a specific rule or logic gap and fixed.

Applications. *Medical diagnosis:* MYCIN identified bacterial infections and recommended antibiotics. *Finance:* credit scoring, fraud detection, investment advice. *Technical support:* guided troubleshooting from predefined rules. *Manufacturing:* process optimisation, quality control and inventory management. Other classic examples are DENDRAL (chemical structure analysis) and XCON (computer configuration).

6.2 Natural Language Processing: Levels of Analysis, Challenges and Applications

Natural language processing (NLP)

NLP is the branch of AI that enables computers to understand, analyse, interpret and generate human language. Processing happens in several levels, each adding a deeper layer of understanding of structure and meaning.

Levels of analysis

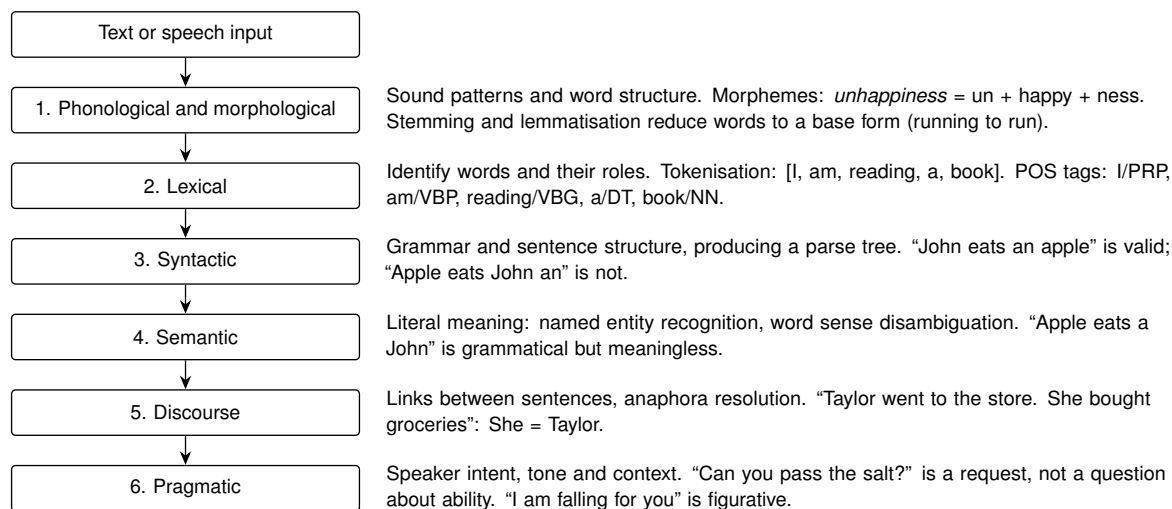
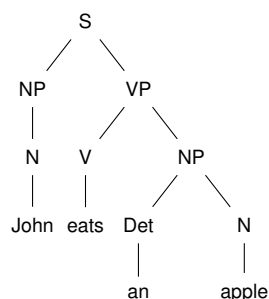


Figure 10: Levels of analysis in NLP, from surface form to intended meaning.



Parse tree of "John eats an apple" at the syntactic level: a sentence (S) is a noun phrase (NP) followed by a verb phrase (VP); the VP is a verb (V) and another NP made of a determiner (Det) and a noun (N). POS tagging and resolving ambiguity ("book" as noun or verb) are the key tasks here.

Figure 11: A parse tree for syntactic analysis.

NLU versus NLG

NLP has two halves. **Natural language understanding (NLU)** maps text to meaning; **natural language generation (NLG)** maps meaning or data to text.

Aspect	NLU	NLG
Direction	Text (or speech) to structured meaning.	Structured data or meaning to text.
Main tasks	Intent detection, entity recognition, sentiment analysis, parsing.	Summarisation, report writing, response generation, output of translation.
Core difficulty	Ambiguity and missing context.	Choosing content, and producing fluent, correct text.
Example	"Book a flight to Pokhara tomorrow" becomes intent = book flight, place = Pokhara, date = tomorrow.	Weather data (rain, 22 degrees) becomes "Expect rain in Kathmandu with a high of 22 degrees."

Challenges in NLP

- **Ambiguity:** the biggest hurdle, at word level ("bank") or structure level: "I saw the man with the telescope" has two readings.
- **Sarcasm and irony:** "Oh great, another flat tire" is negative, but a model may see the word "great" and call it positive.
- **Slang and new words:** language changes faster than dictionaries ("straight fire" means excellent).
- **Domain specific language:** a model trained on Wikipedia may fail on medical or legal text; "bug" means an error to an engineer and an insect to a biologist.

Applications: machine translation, sentiment analysis, chatbots and virtual assistants, text summarisation, speech recognition, question answering, spam filtering.

6.3 Robotics and Computer Vision

Robotics: fundamentals and components

Robotics is the interdisciplinary field (mechanical and electrical engineering, computer science) concerned with the design, construction, operation and use of robots and the computer systems that control them. A **robot** is a reprogrammable, multifunctional manipulator that moves material, parts, tools or devices through variable programmed motions to perform a variety of tasks.

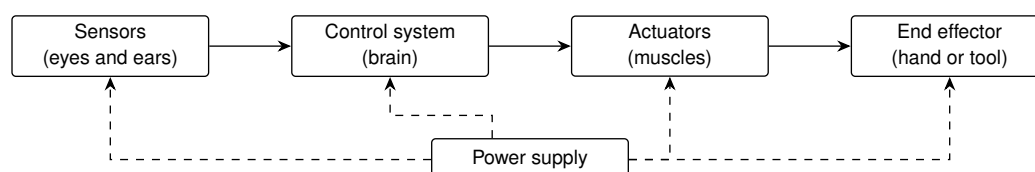


Figure 12: Fundamental components of a robot.

- **Sensors** gather information about the robot's state or surroundings (lidar for distance, joint angle sensors, cameras).
- **Actuators** convert stored energy into movement (DC motors, servo motors, hydraulic cylinders).
- **Control system** processes sensor data and commands actuators, from a small microcontroller to an AI computer on a Mars rover.
- **Power supply:** batteries, tethered cables or solar panels.
- **End effectors** are the tools at the end of an arm: gripper, welding torch, suction cup.

Robot classification (by kinematics, that is how they move):

Type	Description and use
Articulated	Arm like, typically 6 rotary joints (6 degrees of freedom); most common in manufacturing (welding, painting).
SCARA	Selective Compliance Assembly Robot Arm: rigid vertically, flexible horizontally; fast pick and place work.
Cartesian (gantry)	Moves in straight lines along X, Y, Z axes; precise and stable, used in 3D printing and CNC machining.
Mobile	Not fixed in place. AGVs (automated guided vehicles) follow magnetic strips; AMRs (autonomous mobile robots) navigate freely using maps.

Computer graphics, image processing and computer vision

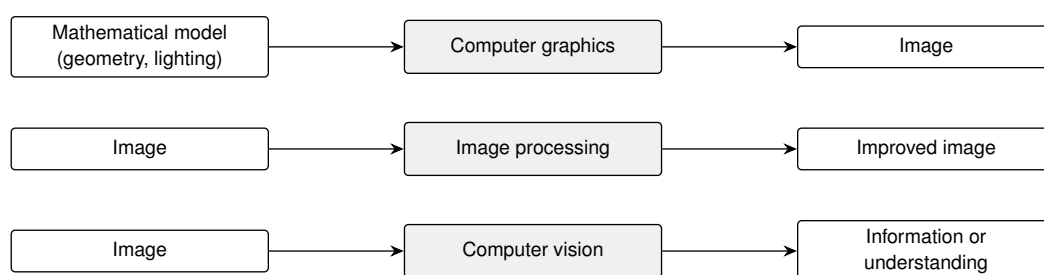


Figure 13: Input and output of the three related fields.

Examples: a game engine rendering a 3D dragon (graphics); adjusting brightness or removing red eye (image processing); a system that outputs “3 cars and 1 pedestrian” from a street photo (computer vision).

Computer vision: core concepts

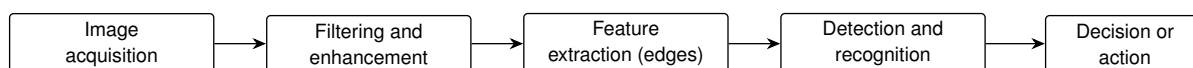


Figure 14: A typical computer vision pipeline.

An image is a matrix of pixel intensities $I(x, y)$ (0 to 255 for an 8 bit grey image).

Image filtering and enhancement. Filtering changes each pixel using its neighbours, by convolution with a small kernel K :

$$(I * K)(x, y) = \sum_i \sum_j I(x - i, y - j) K(i, j).$$

A *Gaussian blur* uses the kernel $G(x, y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}$ to reduce noise. *Histogram equalisation* spreads intensity values to raise contrast so that details in shadows appear. For L grey levels, n pixels in total and n_j pixels at level j , input level k is mapped to

$$s_k = (L - 1) \sum_{j=0}^k \frac{n_j}{n}.$$

Feature extraction (edges). An edge is a sharp change in brightness. Using the horizontal and vertical gradients G_x, G_y (for example from Sobel kernels), the edge strength and direction are

$$|\nabla I| = \sqrt{G_x^2 + G_y^2}, \quad \theta = \tan^{-1}\left(\frac{G_y}{G_x}\right), \quad S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad S_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

The *Canny edge detector* smooths the image, computes gradients, thins edges by non maximum suppression and links them with double thresholds. *SIFT* finds corners and blobs that remain recognisable under rotation and scaling.

Object detection and recognition. *Detection* asks “where is it?” and draws a bounding box; *recognition* asks “what is it?” and assigns a label. Convolutional networks such as YOLO (You Only Look Once) do both in a single pass within milliseconds.

Why deep learning often beats traditional computer vision. (i) It *learns features from data* instead of relying on hand designed ones such as edges and SIFT. (ii) Layers build a hierarchy: edges, then parts, then whole objects. (iii) Very large labelled datasets (for example ImageNet) and GPUs make training practical. (iv) End to end training optimises the whole pipeline for the final task. (v) Learned features are more robust to lighting, pose and background changes.

Applications

Robotics	Description
Industrial automation	The largest sector; handles “3D” jobs (dull, dirty, dangerous): spot welding and painting car bodies around the clock.
Surgical assistance	Systems such as da Vinci let a surgeon operate from a console with very fine, steady movements, giving smaller incisions and faster recovery.
Space exploration	Rovers such as Perseverance navigate, drill and analyse soil on Mars autonomously, surviving radiation and extreme temperature.
Computer vision	Description
Autonomous vehicles	Detect lane lines, read traffic signs, identify pedestrians and brake automatically.
Facial recognition	Phone unlock by mapping face geometry; airport e gates matching a face to a passport photo.
Medical imaging	A “second pair of eyes” on X rays and MRI, highlighting faint tumours or fractures.
Quality control	High speed cameras inspect every item on a line and reject defects such as a missing cap or torn label.

6.4 Sustainable AI Systems

Sustainable AI

Sustainable AI means designing, training, deploying and operating AI systems so as to minimise environmental impact (energy, carbon, water, electronic waste) while maximising social and economic benefit, across the whole AI lifecycle.

AI for sustainability (AI as a tool)	Sustainability of AI (AI's own footprint)
Solve environmental and resource problems: grids, climate, agriculture, conservation.	Reduce the environmental cost of compute, energy, hardware and cooling water.

AI to optimise energy grids, predict climate and improve resource efficiency.

- *Energy grids*: deep models forecast electricity demand to balance supply in real time; predict solar and wind output to smooth intermittency; detect anomalies that signal transformer failure or line loss.
- *Climate change*: downscale coarse climate model output to high resolution regional forecasts; predict hurricane paths and flood risk; estimate carbon flux from satellite and sensor data.
- *Resource efficiency*: vision and sensors for precise irrigation, predictive maintenance to cut material waste, leak detection in water networks, robotic sorting of recyclables.

Energy consumption of large models. Training a multibillion parameter language model can consume hundreds of megawatt hours. Emissions depend on the energy used E and the carbon intensity of the local grid I_{grid} :

$$E_{\text{CO}_2} = E \times I_{\text{grid}}, \quad E \approx N_{\text{GPU}} \times P \times t \times \text{PUE},$$

where P is average power per device, t is run time and PUE (power usage effectiveness) accounts for data centre overhead such as cooling. *Example*: If training uses 500 MWh (= 500,000 kWh) on a grid emitting 0.4 kg CO₂ per kWh, emissions are $500,000 \times 0.4 = 200,000$ kg, that is 200 tonnes of CO₂. Data centres also use large amounts of fresh water for cooling. Training is a one time cost, but inference runs repeatedly at scale and can dominate the lifetime energy use.

Green AI practices. Green AI treats energy and carbon as evaluation metrics alongside accuracy.

- **Quantisation**: lower weight precision from FP32 to INT8 or INT4, cutting memory (INT8 is 4 times smaller than FP32) and energy per operation.
- **Pruning**: remove redundant weights or connections with little loss in accuracy.
- **Knowledge distillation**: train a small student model to mimic a large teacher model.
- **Sparse mixture of experts**: activate only a few sub networks per input.
- **Carbon aware scheduling**: run heavy jobs when and where the grid is cleaner.
- **Efficient cooling**: closed loop liquid or immersion cooling to cut freshwater use.

Chapter 7 Emerging Trends

7.1 Sequence to Sequence Models

Sequence to sequence (Seq2Seq) model

A Seq2Seq model is a deep learning architecture that maps an input sequence of arbitrary length to an output sequence of a different arbitrary length. It is built from an **encoder** and a **decoder**.

Why an encoder and decoder? Ordinary feedforward or convolutional networks map a fixed size input to a fixed size output. Real sequences (language, audio, time series) break this in two ways: input and output have *variable and unmatched lengths* (a 5 word English sentence may become 8 Nepali words), and words may be *aligned out of order* between the two languages. Separating reading the input from writing the output solves both.

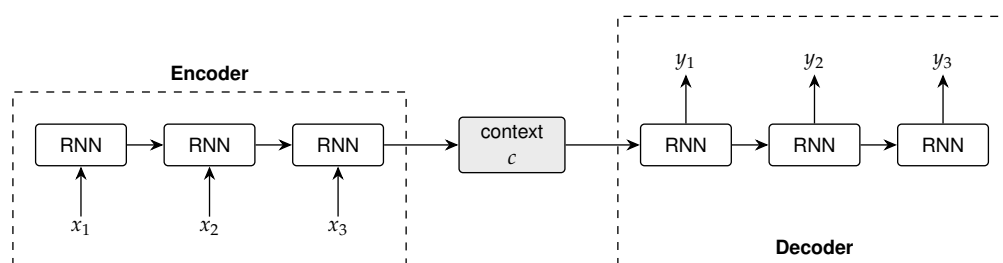


Figure 15: Encoder decoder structure. The decoder emits one token per step, feeding each back in, until $\langle \text{EOS} \rangle$.

- **Encoder:** converts the discrete input tokens into continuous vectors that capture semantic, syntactic and contextual relationships, and summarises them in a context vector: $h_t = f(x_t, h_{t-1})$, $c = h_T$.
- **Decoder:** generates the output token by token, autoregressively, until an end of sequence token $\langle \text{EOS} \rangle$: $s_t = g(s_{t-1}, y_{t-1}, c)$ and $P(y_t | y_{<t}, x) = \text{softmax}(Ws_t + b)$. The whole output probability is $P(y | x) = \prod_{t=1}^{T'} P(y_t | y_{<t}, x)$.
- **Attention:** a single fixed context vector is a bottleneck for long inputs. Attention lets the decoder look at all encoder states, $c_t = \sum_j \alpha_{tj} h_j$ with weights $\alpha_{tj} = \text{softmax}_j(e_{tj})$. Transformers use self attention throughout: $\text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V$.

Evolution: RNN and LSTM encoder decoders (2014 to 2015), then attention (2015 to 2017), then Transformers (2017 to present), with state space models such as Mamba (2021 onwards) as newer alternatives.

Benefits. Handles variable length input and output; learns end to end from paired examples without hand written rules; one general framework serves many tasks; attention and Transformers capture long range context and train in parallel.

Applications. Machine translation (for example English to Nepali), abstractive text summarisation, chatbots and conversational AI, speech recognition (audio to text), speech synthesis (text to audio), question answering, code generation from instructions, image captioning (CNN encoder with an RNN or Transformer decoder).

7.2 Federated Learning

Federated learning (FL)

FL is a distributed machine learning approach in which a model is trained across many decentralised devices or servers that hold local data, *without the raw data ever leaving its source*. Only model updates are shared.

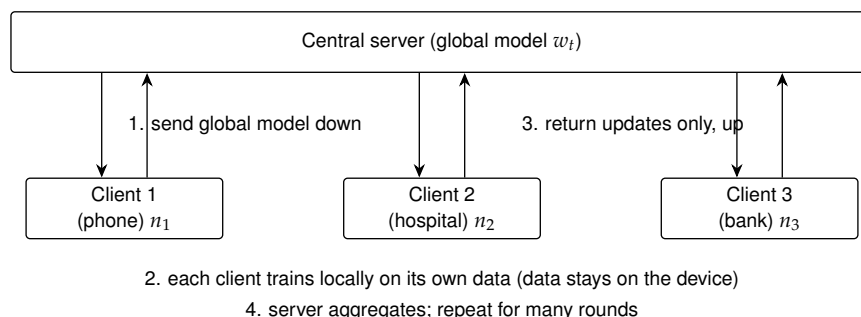


Figure 16: One communication round of federated learning.

How it works. (1) The server initialises a global model and sends it to a subset of clients. (2) Each client trains it for a few epochs on local data. (3) Each client returns only the updated parameters. (4) The server aggregates them, usually by a data size weighted average (*FedAvg*):

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_{t+1}^k, \quad n = \sum_{k=1}^K n_k.$$

(5) The cycle repeats until the global model converges. *Example:* Three clients with $n = 100, 200, 700$ and local weights $0.2, 0.5, 0.4$ give $w = 0.1(0.2) + 0.2(0.5) + 0.7(0.4) = 0.40$.

Benefits. Privacy and confidentiality (raw data stays local); help with regulations such as GDPR and HIPAA; lower bandwidth (updates are much smaller than datasets); access to data that could never be centralised (cross hospital or cross bank); personalisation by local fine tuning; continuous on device learning.

Applications. Gboard next word prediction from millions of phones; hospitals jointly training tumour detection models; banks jointly detecting fraud; smart speakers and wearables; fleets of vehicles improving perception models. *Challenges:* non identically distributed client data, communication rounds, device differences, and residual leakage from updates (often reduced with differential privacy).

Machine learning versus federated learning.

Aspect	Traditional (centralised) ML	Federated learning
Data location	All data collected and stored on one server.	Data stays on the clients.
What is sent	Raw data to the server.	Model updates (weights or gradients) only.
Training site	Central server or data centre.	On every client, then aggregated centrally.
Privacy	Higher risk; raw data is exposed and centralised.	Better privacy by design.
Communication	One time upload of a large dataset.	Many rounds of small updates.
Data distribution	Usually pooled and shuffled (roughly identical).	Non identical, uneven across clients.
Compliance	Harder under GDPR and HIPAA.	Easier, since data does not move.

7.3 Edge AI

Edge AI

Edge AI runs AI models directly on local (edge) devices such as smartphones, sensors, cameras, drones and microcontrollers, instead of sending data to a distant cloud server and waiting for the reply. Computation happens where the data is generated.

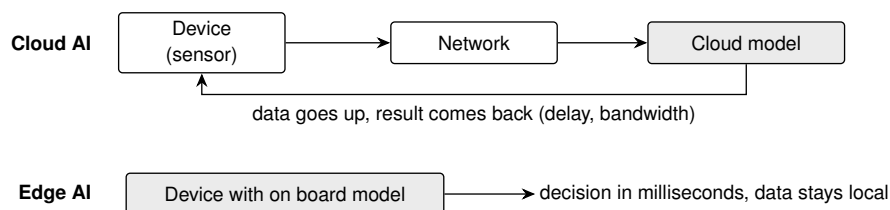


Figure 17: Cloud AI versus Edge AI.

Key enablers. *Model compression* to fit constrained hardware: quantisation (32 bit floats to 8 bit integers), pruning (removing low importance weights) and knowledge distillation (a small student mimics a large teacher). *Specialised hardware:* NPUs, TPUs, mobile GPUs, accelerators such as Google Coral and NVIDIA Jetson. *Lightweight frameworks:* TensorFlow Lite, ONNX Runtime, PyTorch Mobile.

Benefits. Low latency (milliseconds, vital for driving and industrial safety); lower bandwidth and cost; better privacy and security (raw data processed and discarded locally); works offline or with poor connectivity; scales by spreading computation; saves energy spent transmitting raw streams.

Applications. On device face unlock and wake word detection on smartphones; real time object and lane detection in autonomous vehicles; defect detection and predictive maintenance in factories; smart cameras that detect people without streaming video; heart rate and fall detection on wearables; crop and pest monitoring by drones in remote fields (relevant to rural Nepal); smart shelves in retail.

7.4 Ethics and AI: Responsible AI

Responsible AI

Responsible AI is a governance framework and ethical approach for designing, building and deploying AI systems that are safe, trustworthy and beneficial to society. It moves from “what can we build?” to “what should we build?”, respecting human rights and democratic values.

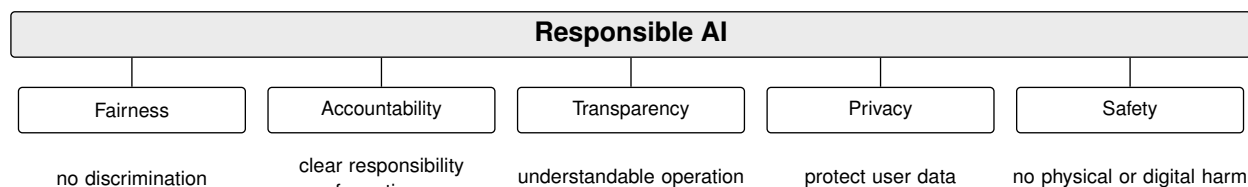


Figure 18: The five core pillars of Responsible AI.

Framework for developing and deploying AI

Ethical checks are embedded at every stage of the lifecycle.

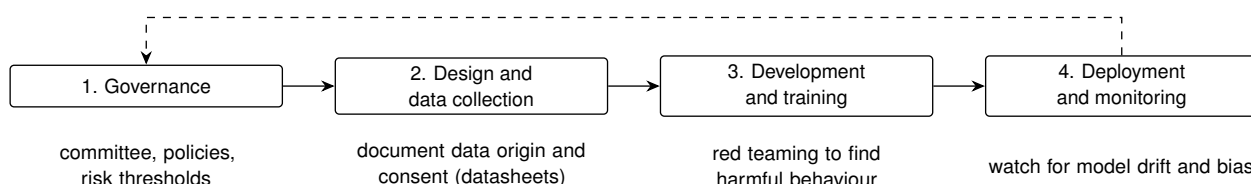


Figure 19: Responsible AI lifecycle.

Bias and fairness

Bias is a systematic error that produces unfair outcomes, often favouring one group. *Historical bias*: training data reflects past prejudice. *Sampling bias*: data does not represent the real population (a face recognition system trained mostly on light skinned faces fails on dark skinned faces). *Proxy bias*: a care allocation model used healthcare cost as a proxy for need and underestimated the needs of groups with less access to care.

Fairness is the deliberate correction of bias, and has several definitions. *Group fairness* requires similar outcome rates across groups; for example demographic parity: $P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = b)$ for protected attribute A . *Individual fairness* requires that similar individuals receive similar outcomes. Mitigation includes representative data, bias audits and regular monitoring.

Explainable AI (XAI)

XAI is a set of methods that make a model’s outputs understandable and trustworthy to humans. It is a part of the fairness, accountability and transparency (FAT) paradigm and is especially needed for deep learning.

- **Principles:** transparency, interpretability, accountability. **Benefits:** better decisions, more trust and acceptance, lower risk and liability.
- **Approaches:** feature importance, attribution and visualisation. **Techniques:** *LIME* (fits a simple local model around one prediction), *SHAP* (assigns each feature a Shapley value, its average marginal contribution to the prediction), *ELI5*.

Privacy risks and AI security threats

Privacy risks. Models can memorise training data and leak it. In a *membership inference attack* an attacker learns whether a person's record was in the training set (which can reveal, for example, a diagnosis). In *model inversion* the attacker reconstructs training data, such as a face, from model outputs or confidence scores. Simply masking names is not enough because records can be cross referenced. *Differential privacy* adds calibrated noise so that any single individual's presence barely changes the output: $P[M(D) \in S] \leq e^\epsilon P[M(D') \in S]$ for datasets D, D' differing in one person. Federated learning keeps raw data on the device.

Security threats and defences.

Threat	Description and example	Defence
Adversarial (evasion) attack	Small crafted change to an input, $x' = x + \delta$ with $\ \delta\ \leq \epsilon$, makes the model err: a sticker makes a stop sign read as a speed limit sign.	Adversarial training (train on attacked examples); input sanitisation (for example slight blurring).
Data poisoning	Malicious data injected during training: Microsoft's Tay chatbot turned offensive after users fed it abusive tweets.	Outlier detection on training data; provenance checks.
Model drift or erratic behaviour	Performance degrades after deployment.	Continuous model monitoring, automatic shutdown for inspection.

A "defence in depth" approach combines these layers.

Ethical dilemmas and regulatory frameworks

- **Job displacement:** AI replaces cognitive as well as physical labour, creating risk of technological unemployment.
- **Misuse:** deepfakes and disinformation threaten reputation and politics, and create the "liar's dividend" where real evidence is doubted.
- **Bias and surveillance:** predictive policing trained on biased arrest data, and facial recognition that removes anonymity in public.

Regulatory approaches (for the policy review activity). The *EU AI Act* is a risk based law: unacceptable risk (for example social scoring) is banned, and high risk uses (surgery, hiring, border control) face strict transparency and human oversight. The *US* approach under a 2023 Executive Order stressed standards and national security, requiring developers of powerful dual use foundation models to share red team results with government (this order was later rescinded in 2025). *China* regulates generative AI with content rules and provider responsibility. Common policy themes are trust and transparency, ethical principles, accountability, bias mitigation, safety and security, continuous adaptation and legal compliance.

Knowledge Check: 40 MCQs

Choose the single best answer. Answers are given at the end. Questions 1 to 16 cover Chapter 1, 17 to 30 cover Chapter 6, and 31 to 40 cover Chapter 7.

Chapter 1: Introduction

1. Which event is widely regarded as the birth of AI as a field?
(A) The building of ENIAC (1946) (B) The Dartmouth conference (1956)
(C) IBM Deep Blue's chess victory (1997) (D) The publication of the Transformer paper (2017)
2. Which capability is NOT required to pass the standard, text only Turing test?
(A) Natural language processing (B) Knowledge representation
(C) Computer vision (D) Machine learning
3. A CAPTCHA is an example of the:
(A) Reverse Turing test (B) Total Turing test
(C) Cognitive modelling approach (D) Laws of thought approach
4. Which approach to AI is considered the most appropriate because it is measurable and handles uncertainty?
(A) Thinking humanly (B) Thinking rationally
(C) Acting humanly (D) Acting rationally
5. The first AI winter occurred approximately during:
(A) 1956 to 1962 (B) 1974 to 1980
(C) 1987 to 1993 (D) 1997 to 2003
6. In the DIKW pyramid, the correct order from bottom to top is:
(A) Data, Knowledge, Information, Wisdom (B) Information, Data, Wisdom, Knowledge
(C) Data, Information, Knowledge, Wisdom (D) Knowledge, Data, Information, Wisdom
7. Data becomes information when it is:
(A) organised and interpreted in a meaningful way (B) stored on a larger disk
(C) encrypted for security (D) compressed to save space
8. The human brain has roughly how many neurons?
(A) 86 million (B) 8.6 billion
(C) 860 billion (D) 86 billion
9. Which brain region is associated with logic, language and planning, and inspired deep learning?
(A) Cerebellum (B) Neocortex
(C) Limbic system (D) Brain stem
10. In the PEAS framework, the cleanliness of the floor for a vacuum cleaner agent belongs to the:
(A) Environment (B) Actuators
(C) Performance measure (D) Sensors

11. Which agent acts only on the current percept and has no memory of past percepts?
(A) Simple reflex agent (B) Model based reflex agent
(C) Goal based agent (D) Utility based agent
12. An agent that chooses among competing goals using a measure of how desirable each outcome is, is called a:
(A) Simple reflex agent (B) Goal based agent
(C) Model based reflex agent (D) Utility based agent
13. In a learning agent, the component that suggests exploratory actions is the:
(A) Critic (B) Problem generator
(C) Performance element (D) Sensor
14. Which best distinguishes agentic AI from earlier generative AI?
(A) It only generates longer text (B) It needs no training data
(C) It pursues goals by planning, using tools and self correcting (D) It never interacts with humans
15. Which is an example of autonomous intelligence?
(A) NASA's Perseverance rover navigating on Mars (B) A spreadsheet macro
(C) A pocket calculator (D) A static web page
16. A rational agent selects the action that:
(A) always minimises computation time (B) copies human behaviour exactly
(C) always chooses the first available action (D) maximises its expected performance measure given its percepts

Chapter 6: AI Applications

17. Which expert system component shows how a conclusion was reached?
(A) Knowledge acquisition module (B) Working memory
(C) Explanation module (D) User interface
18. Backward chaining in an inference engine is:
(A) goal driven, starting from a hypothesis (B) data driven, starting from known facts
(C) a random search of the rules (D) a training method for neural networks
19. The expert system development phase in which the knowledge engineer interviews the domain expert is:
(A) Implementation (B) Testing and validation
(C) Problem identification (D) Conceptualisation
20. A key limitation of expert systems is that they:
(A) give different answers to the same facts (B) cannot handle situations outside their rules
(C) forget facts after every use (D) cannot run more than once a day
21. Tokenisation and part of speech tagging belong to which NLP level?
(A) Lexical analysis (B) Pragmatic analysis
(C) Discourse analysis (D) Semantic analysis

22. "Apple eats a John" is grammatical but meaningless. Which level detects the problem?
- (A) Lexical analysis (B) Syntactic analysis
(C) Semantic analysis (D) Phonological analysis
23. Interpreting "Can you pass the salt?" as a polite request is done at the ... level.
- (A) syntactic (B) pragmatic
(C) morphological (D) lexical
24. Producing a sentence from weather data is an example of:
- (A) natural language understanding (B) named entity recognition
(C) stemming (D) natural language generation
25. Which robot type suits fast pick and place work, being rigid vertically and flexible horizontally?
- (A) Cartesian gantry (B) Mobile AGV
(C) SCARA (D) Legged humanoid
26. Which field takes an image as input and outputs information or understanding?
- (A) Computer vision (B) Computer graphics
(C) Image processing (D) Digital printing
27. Given gradients G_x and G_y , the edge strength is:
- (A) $G_x + G_y$ (B) $G_x G_y$
(C) G_x / G_y^2 (D) $\sqrt{G_x^2 + G_y^2}$
28. YOLO is used in computer vision to:
- (A) blur an image with a Gaussian kernel (B) detect and recognise objects in a single pass
(C) render 3D models (D) compress video files
29. Quantisation in Green AI means:
- (A) adding more layers to the model (B) increasing the batch size
(C) reducing numerical precision of weights, for example FP32 to INT8 (D) building larger data centres
30. Training uses 200 MWh on a grid emitting 0.5 kg CO₂ per kWh. The emissions are:
- (A) 100 tonnes (B) 10 tonnes
(C) 400 tonnes (D) 1000 tonnes

Chapter 7: Emerging Trends

31. In a Seq2Seq model the decoder keeps generating tokens until it produces:
- (A) a padding token (B) the first encoder state
(C) a blank pixel (D) the end of sequence token
32. Which architecture introduced in 2017 replaced recurrent layers in most sequence models?
- (A) LeNet (B) Transformer
(C) Perceptron (D) Hopfield network
33. In federated learning, each client sends the server:
- (A) model updates (weights or gradients) (B) its raw training data
(C) user passwords (D) a full copy of its database

34. Three clients hold $n = 100, 200, 700$ samples with local weights 0.2, 0.5, 0.4. FedAvg gives:
- (A) 0.37 (B) 0.20
(C) 0.40 (D) 0.70
35. The main benefit of Edge AI for autonomous driving is:
- (A) unlimited compute power (B) low latency, local decisions
(C) no need for sensors (D) no need for a trained model
36. Training a small student model to mimic a large teacher model is called:
- (A) pruning (B) quantisation
(C) federated averaging (D) knowledge distillation
37. Which is NOT a core pillar of Responsible AI?
- (A) Fairness (B) Accountability
(C) Profit maximisation (D) Transparency
38. A sticker on a stop sign that makes a vision model read a speed limit sign is:
- (A) an adversarial (evasion) attack (B) data poisoning
(C) a membership inference attack (D) model drift
39. Microsoft's Tay chatbot turning offensive after users fed it abusive tweets illustrates:
- (A) adversarial evasion (B) differential privacy
(C) edge inference (D) data poisoning
40. Under the EU AI Act, social scoring systems are classified as:
- (A) minimal risk (B) unacceptable risk (banned)
(C) high risk, allowed with audits (D) limited risk

Answer Key

1: B	2: C	3: A	4: D	5: B	6: C	7: A	8: D	9: B	10: C
11: A	12: D	13: B	14: C	15: A	16: D	17: C	18: A	19: D	20: B
21: A	22: C	23: B	24: D	25: C	26: A	27: D	28: B	29: C	30: A
31: D	32: B	33: A	34: C	35: B	36: D	37: C	38: A	39: D	40: B
